VitalTrack EEG User Manual

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Newert Co., Ltd.

NEWERT

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1. Getting started

1.1 Product overview

VitalTrack EEG is a wireless device that measures the electroencephalogram (EEG) through electrodes placed on the scalp. The device connects to the VitalTrack app (PC and mobile) over Bluetooth LE; the app shows the waveforms in real time and records the data to files.

- Measures up to 8 EEG channels at a sampling rate of 250 Hz.
- Records head movement with a built-in inertial measurement unit (IMU).
- Stores recordings as CSV files on the device you use. No data is sent to an external server.

1.2 Package contents

Item	Quantity
VitalTrack EEG device (battery connected)	1
Electrode cap (20 channels)	1
Electrode set (10 electrodes)	1
Earlobe clips (REF, BIAS)	2
USB-C cable	1

Prepare the following separately: conductive gel or saline (whichever suits the electrode cap), a needleless syringe, and alcohol swabs.

1.3 Safety and precautions

- The device and app are intended for **research and wellness use**. Do not use them for medical diagnosis or treatment.
- Do not place electrodes on wounds, inflamed skin, or skin conditions.
- Do not attach electrodes to the body while the device is connected to a PC or charger by USB. Remove the device from the body before charging.
- If you feel skin irritation or discomfort during a measurement, remove the electrodes immediately.
- Use only the supplied battery. Do not puncture, crush, or heat the battery.
- If the battery swells, gets hot, or smells, stop using the device immediately and unplug the USB cable.
- Keep water, gel, and saline away from the device. Do not leave it in hot, humid places or in direct sunlight.

1.4 System requirements

Platform	Requirement
PC	Windows 10 or later, Bluetooth LE
Android	Android 7.0 or later
iOS	iPhone (distributed through TestFlight)

Screens in this manual are from Android. The PC and iOS apps have the same layout; differences are noted separately.

2. Installing the app

2.1 PC (Windows)

1. Extract the provided archive (`VitalTrack-Windows-….zip`) to a folder of your choice.
2. Run `VitalTrack.exe` in that folder.
3. If "Windows protected your PC" appears, select **More info** → **Run anyway**.
4. Make sure Bluetooth is turned on in Windows Settings.

2.2 Android

1. Install VitalTrack from the provided test link.
2. On first launch the app asks for permissions. Select **Allow** for each.
   - **Nearby devices**: required to find and connect to the device.
   - **Location** (Android 11 and earlier): used only for the Bluetooth device scan.
   - **Notifications**: shows that a recording is in progress.
   - **Exclude from battery optimization**: keeps recording when the screen is off.

> **iOS**
>
> 1. Install the TestFlight app, then install VitalTrack from the provided invitation link.
> 2. Allow Bluetooth access on first launch. iOS does not ask for location access.

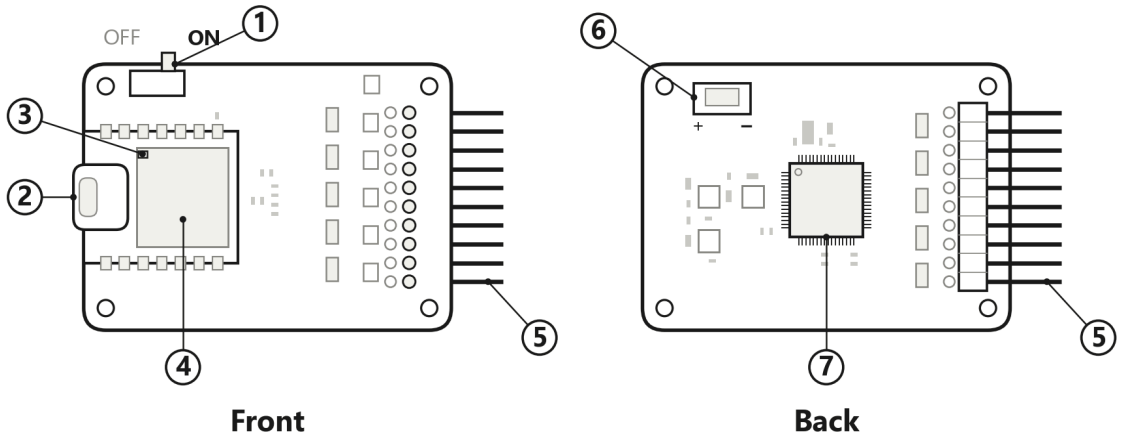
2.3 Test version expiry

A test version can be used until a set date. After that date the app shows "The test period has ended" at launch; install a new version. Recordings already saved are kept.

3. Hardware

The device is a single circuit board. The power switch, USB-C port and wireless module are on the front; the battery connector and the signal acquisition circuit are on the back. The electrode connector is on one edge of the board.

3.1 Parts



Parts of the device

Name	Description
① Power switch	Slide towards ON to turn the device on and towards OFF to turn it off.
② USB-C port	Charges the battery.
③ Status LED	Shows the connection and charging state by colour (see 3.2).
④ Wireless module	Contains the Bluetooth LE radio and the motion sensor (IMU).
⑤ Electrode connector	10-pin connector for the REF, CH1–CH8 and BIAS electrodes (see 3.3).
⑥ Battery connector	2-pin connector for the battery. The polarity (+, −) is marked on the board.
⑦ Signal acquisition	Amplifies the EEG and converts it to a digital signal.

> The circuit board of the device is exposed. Hold the board by its edges, and do not touch the circuit or the connector pins during a measurement. Do not place the device on metal objects.

3.2 Power and charging

Turning the device on and off

- Slide the power switch towards ON. The status LED blinks blue and the device can be found in the app.

- When you are done, slide the switch towards OFF. Turning the device on again restarts it, so connect it again in the app.

Charging

1. Remove the device from the body.
2. Connect the device to a USB port of a PC or to a USB charger with the USB-C cable. The device charges even when the power switch is OFF.
3. When the status LED is solid green, charging is complete.

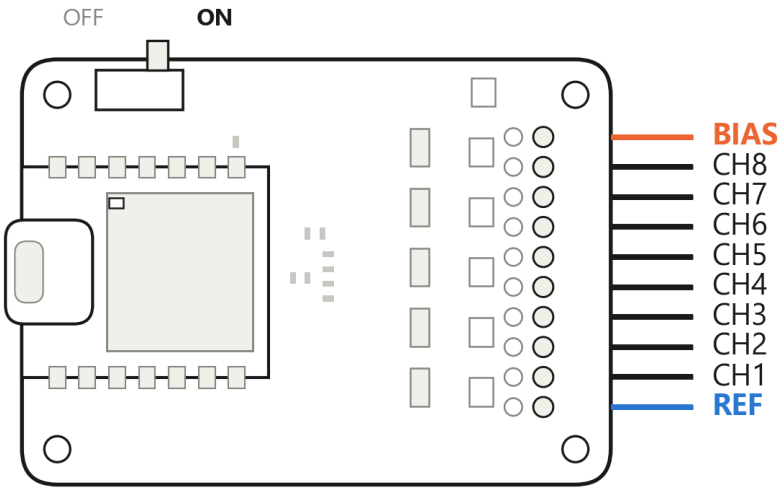
The charging current is low (50 mA), so an empty battery can take more than 10 hours to charge. Charge the device the day before a measurement.

Status LED

LED	Meaning
Blinking blue	Waiting for a connection. The device can be found in the app.
Solid blue	Connected to the app.
Solid red	Running on the battery.
Blinking red, yellow or green	Charging. The colour shows the battery level (red low, yellow medium, green high).
Solid green	Charging is complete.

The app shows the battery level and power state of the connected device (USB · Charging, USB · Charged). When the level drops below the set value, the app shows a warning (see 10.4 Battery warnings). The level is estimated from the battery voltage, so it can stay below 100% right after charging. Use the status LED to check that charging is complete.

3.3 Electrode connector pinout



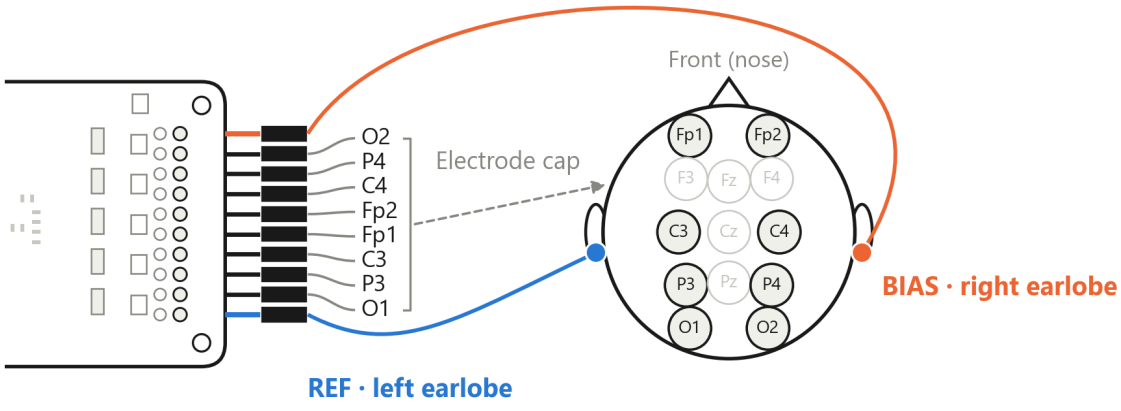
Electrode connector pinout (front side)

The electrode connector has 10 pins at a 2.54 mm pitch. With the front side up and the connector on the right, the pins are in this order, starting from the power switch side.

Order	Pin	Electrode	Circuit name
1	BIAS	Bias electrode (GND on the cap)	BIAS
2	CH8	Channel 8 electrode	IN8N
3–8	CH7 – CH2	Channel 7 – 2 electrodes	IN7N – IN2N
9	CH1	Channel 1 electrode	IN1N
10	REF	Reference electrode	SRB2

The row of holes on the inner side of the connector pins is not used.

3.4 Connecting the electrodes



Connecting the electrodes

1. Turn the device off.
2. Plug the REF earlobe clip onto the **REF** pin and the BIAS earlobe clip onto the **BIAS** pin.
3. Plug the electrode wires of the cap onto the channel pins you use (CH1–CH8). Push each connector straight onto its pin as far as it goes.
4. In the Prepare screen of the app, assign the electrode position plugged into each channel (see 6.2).

Example wiring for the whole scalp (recommended layout in 4.1):

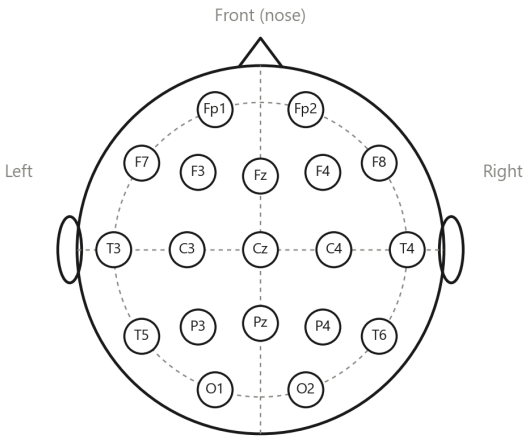
Channel	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
Electrode position	O1	P3	C3	Fp1	Fp2	C4	P4	O2

- The pin an electrode wire is plugged into decides its channel. If you change the wiring, change the channel assignment in the Prepare screen too.
- Leave unused channel pins empty, and leave them out of the channel layout in the Prepare screen (see 4.5).
- To unplug, hold the connector body, not the wire, and pull it straight off. Take care not to bend the pins.

4. Applying the electrodes

4.1 Electrode positions

VitalTrack supports 19 positions of the international 10-20 system. Choose up to 8 of them.



Supported electrode positions (international 10-20 system)

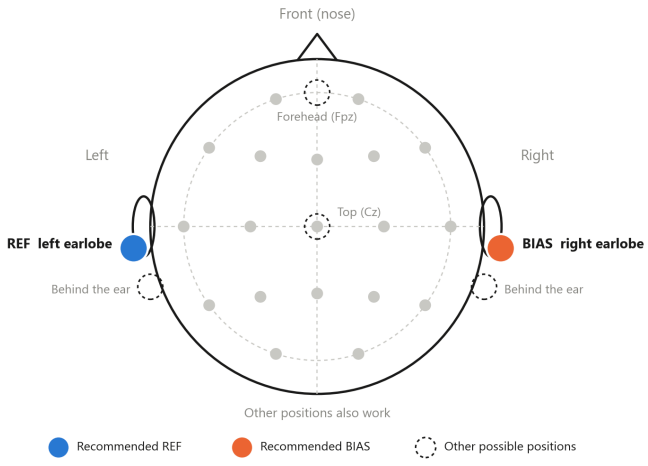
Recommended layouts:

- Occipital alpha: O1, O2, P3, P4
- Whole scalp: Fp1, Fp2, C3, C4, P3, P4, O1, O2

4.2 Reference (REF) and bias (BIAS) electrodes

Every channel is measured as the potential difference to the **reference electrode (REF)**. The **bias electrode (BIAS)** is a driven ground electrode that reduces common-mode noise; connect it to the electrode marked **GND** on the cap.

Recommended: REF on the left earlobe, BIAS on the right earlobe (earlobe clips)



REF and BIAS positions

The following positions can also be used.

Position	REF	BIAS	Notes
Both earlobes (recommended)	Left earlobe	Right earlobe	Earlobe clips are easy to attach and give stable contact.
Mastoids (behind the ears)	Left mastoid	Right mastoid	Needs adhesive electrodes; clenching the jaw can add muscle (EMG) noise.
Vertex and forehead	Vertex (Cz)	Forehead (Fpz)	Needs enough gel through the hair; sensitive to head movement.

The contact of REF and BIAS affects the signal quality of every channel. Check these two electrodes first before measuring.

4.3 Skin preparation and gel

1. Wipe each electrode site lightly with an alcohol swab and let it dry.
2. Part the hair so that the electrode touches the scalp directly.
3. Apply enough conductive gel or paste and press the electrode gently in place. For a saline electrode cap, fill each electrode with saline using a needleless syringe.

4.4 Attaching and checking

1. Attach REF and BIAS (GND) first.

2. Attach the electrodes of the channels you use.
3. Route the cables so that they do not pull or swing.
4. Start **Signal preview** in the app and check that the contact state of each channel shows **Good**. Signal preview is on the Measure tab on mobile and on the Prepare screen on PC (see 6.5 Pre-recording checks).

4.5 Unused channels

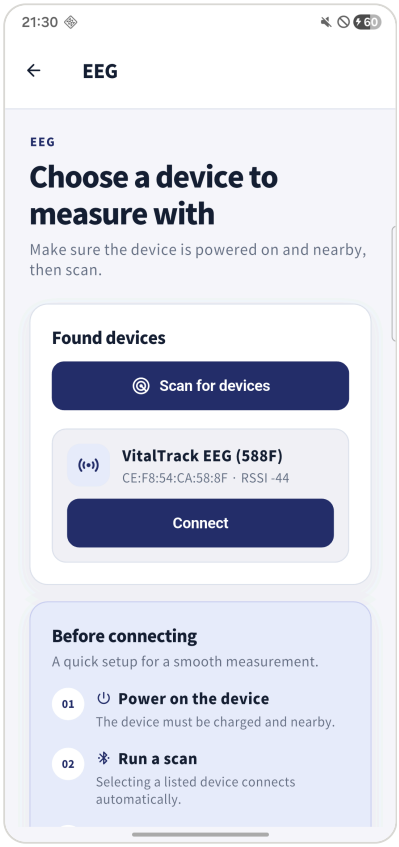
Do not include channels without an electrode in the channel layout. Channels that are not in the layout are switched off on the device, so their noise does not affect the other channels.

5. Connecting the device

5.1 Scanning and connecting

1. Turn on the device.
2. On the first screen of the app, select **EEG**.
3. Select **Scan for devices**. Only devices named "VitalTrack EEG" are shown.
4. Select **Connect** for your device. When connected, the Prepare screen opens.

If the device is not found, see 11.1.



Device scan

5.2 Connection state and battery

The name of the connected device (for example VitalTrack EEG (588F)) and its battery level are shown at the top of the screen.

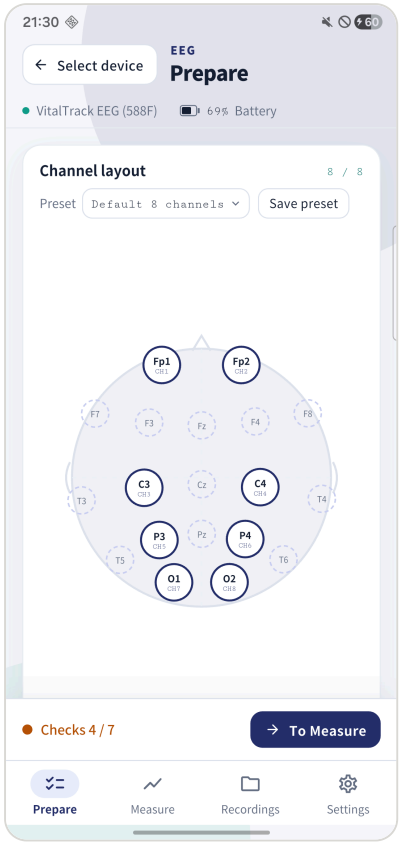
State	Meaning
Streaming	Data is being received from the device.
Reconnecting	The connection was lost; the app is reconnecting automatically.
No data	The connection is up but no data arrives. Check the distance to the device.
USB · Charging / Charged	The device is connected to USB power.

5.3 When the connection is lost

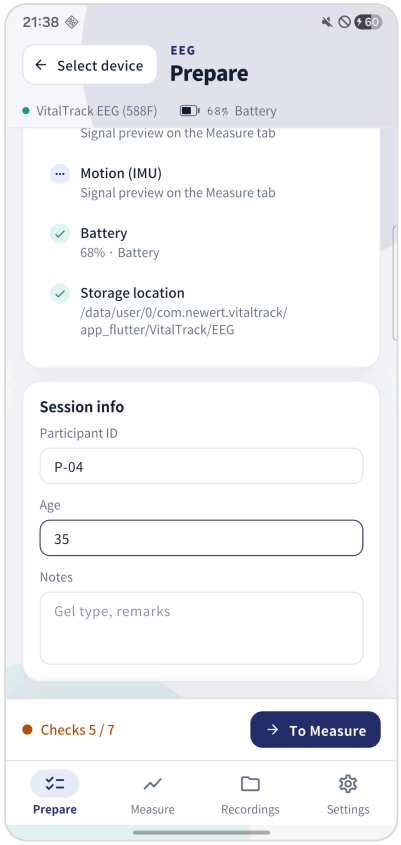
When the connection is lost, the app reconnects automatically and shows the number of attempts. If this happens during a recording, the data continues in the same file. The interrupted period is shown on the timeline of the Recordings screen.

6. Prepare screen

The Prepare screen is where you set the channel layout, enter the session info, and check that everything is ready. The tabs at the bottom switch between **Prepare · Measure · Recordings · Settings**.



Prepare screen



Session info

6.1 Selecting electrode positions

Select the positions you use on the head diagram, up to 8. Select a position again to clear it.

6.2 Assigning input channels

When you select a position, a picker asks which input channel (CH1–CH8) its electrode is connected to. Choose the channel the electrode is actually plugged into on the cable. Channels already assigned to another position are marked "in use".

> If a channel is assigned wrongly, the waveform and the recording carry the wrong position name. Use the cable table in 3.4 to assign the channels correctly.

6.3 Saving and loading presets

Save a layout you use often with **Save preset** and load it later from the **Preset** list.

6.4 Session info

Field	Description
Participant ID	Required. Shown in the file and in the list of recordings.
Age	Required.
Notes	Optional. Recording the measurement conditions, such as the gel type and the REF and BIAS positions, helps later comparisons.

A recording cannot start until the participant ID and age are entered.

6.5 Pre-recording checks

Check the readiness of the measurement in **Pre-recording checks**.

Item	What it checks
Connect device	Whether the device is connected
Channel layout	Number of channels in the layout and the preset
Session info	Whether the participant ID and age are entered
Electrode contact	Number of channels with Good contact (judged while Signal preview runs)
Motion (IMU)	State of the inertial measurement unit
Battery	Battery level of the device
Storage location	Folder where the recording is saved

7. Measure screen

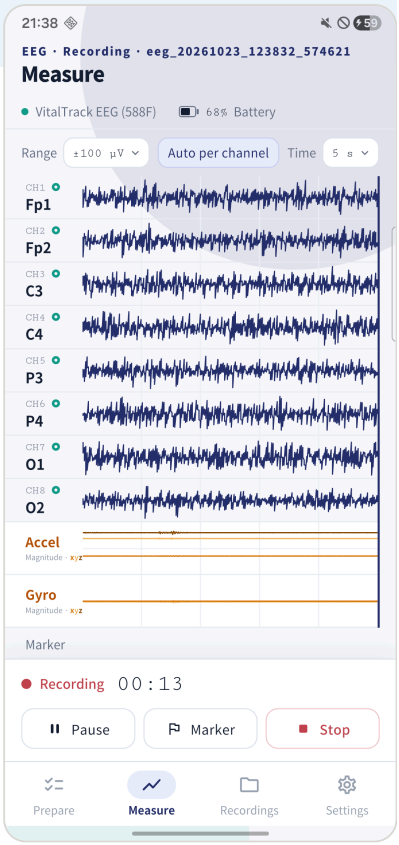
7.1 Layout

- Top: device name, battery, display settings (range and time)
- Middle: waveform of each channel, acceleration and angular velocity, markers
- Bottom: recording state, **Pause**, **Start recording / Stop**

7.2 Waveform display

Control	Description
Range	Sets the vertical range of the waveforms (for example ±100 µV).
Auto per channel	Fits the vertical range of each channel automatically.

Control	Description
Time	Sets the time span shown on screen (seconds).
Pause / Resume	Freezes the display only; recording continues. While paused, swipe sideways to view earlier waveforms.
Signal preview	Shows the waveforms without recording. Preview data is not saved to a file.



Measure screen (recording)

The waveforms on screen pass through a display filter: 5–40 Hz band-pass, mains noise removal (50/60 Hz), and re-referencing to the common (median) reference. The recording file keeps both the raw and the cleaned signal (see 9.2).

7.3 Electrode contact

The colour next to each channel name shows its electrode contact. The state is updated every second from the signal.

State	Meaning	Action
Good	Signal quality is good.	Keep the current setup.
Weak	Mains or other noise is high.	Reduce movement, add gel, or press the electrode in place again.
Not attached	The electrode does not appear to touch the skin.	Check the electrode and cable connection.
Waiting	Not enough signal yet to judge.	Wait a moment.

> The contact state is estimated by software from the signal. It is not a measured electrode impedance.

If several channels turn Weak or Not attached at the same time, check the **REF or BIAS electrode** before the individual electrodes.

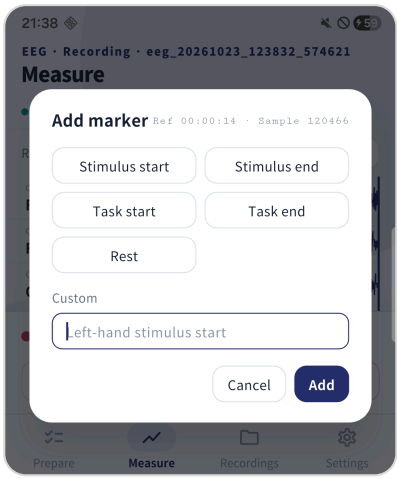
7.4 Motion

The Accel and Gyro lanes show head movement. When movement exceeds the set thresholds (see 10.2), the period is marked as **motion time** and automatic markers are written to the recording file.

7.5 Markers

A marker labels a moment in the recording with an event name.

Type	Description
Manual marker	Select **Marker** and enter or choose an event name. On PC, the **M** key also opens it.
Quick marker	Adds a predefined name (for example Stimulus start, Stimulus end, Rest) in one tap. Names are changed in Settings (see 10.5).
Automatic marker	Added automatically at the start and end of motion time.



Add marker

7.6 Band power and topography

- **Band power**: shows the relative power (%) of the delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and gamma (30–45 Hz) bands for each channel, computed every second over the last 8 seconds.
- **View Topography · PSD**: shows the scalp distribution of the band power and the power spectral density (PSD) of each channel. The distribution is drawn when at least 4 channels are Good.
- Channels judged Not attached are excluded from the analysis.

7.7 Starting and stopping a recording

1. Complete the checks on the Prepare screen.
2. Select **Start recording**. "Recording" appears.
3. When the measurement is finished, select **Stop**. When the file is saved, "Recording complete" appears.

> Other tabs cannot be opened during a recording. Stop the recording first.

8. Recordings screen

8.1 List and search

Saved recordings are listed by date. Search by participant, age, or file name.

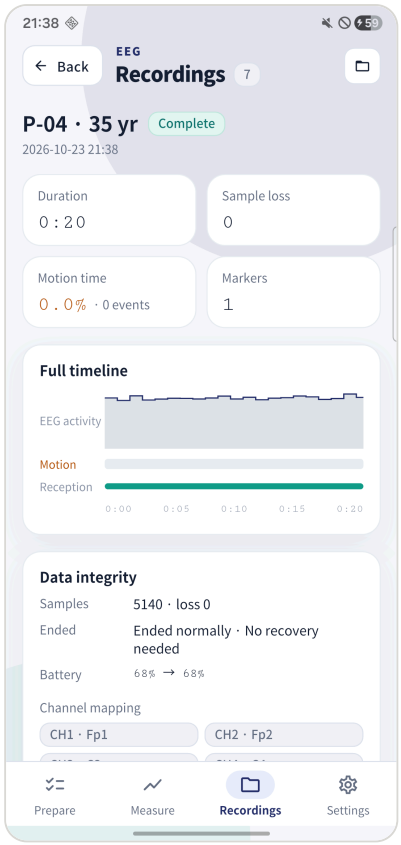
8.2 Recording details

Select a recording to see its details.

- **Full timeline**: EEG activity, motion, and reception over time. Interrupted connections are shown here.
- Summary of **sample loss, motion time, and markers**



Recordings



Recording details

- **Data integrity**: number of saved samples, whether the recording ended normally, device, battery
- **Channel mapping**: the electrode position assigned to each channel
- **Notes**: notes entered on the Prepare screen

8.3 Deleting recordings and opening the folder

- **Delete**: deletes a recording. With **Select**, several recordings can be deleted at once. Deleted recordings cannot be restored.
- **Open folder**: opens the folder that holds the recording files (see 9.1).
- If the app stops unexpectedly during a recording, the recording is marked "Incomplete". **Recover verified samples** restores the correctly saved part to a new file.

9. Managing files

9.1 Storage location

Platform	Location	How to open
PC (Windows)	`Documents\VitalTrack\EEG`	**Open folder** on the Recordings screen or in Settings

Platform	Location	How to open
Android	`VitalTrack/EEG` in the app's storage	**Open folder** opens the VitalTrack folder in the Files app.

> **iOS** Recordings are stored in the Files app under **On My iPhone** → **VitalTrack** → **EEG**. On a Mac, they are also available through File Sharing in Finder.

9.2 Files of a recording

Each recording creates three files.

File	Content
`eeg_date_time_number.csv`	Measurement data, 250 rows per second: raw signal, cleaned signal, inertial data, markers
`eeg_date_time_number_band.csv`	Band power, one row per second: relative band power (%) of each channel
`eeg_date_time_number.csv.session.json`	Session information: participant, channel layout, device, battery, recording summary

The three files share the same name prefix; keep them together when you move or archive them. Appendix A defines each column.

> During a recording the file name ends with `.partial`; the suffix is removed when the recording stops normally.

9.3 Copying to a PC and backing up

- **Android**: connect the phone to a PC by USB, or select the files in the VitalTrack folder of the Files app and share them (email, cloud storage, and so on).
- **PC**: copy the whole `Documents\VitalTrack\EEG` folder to back it up.

> **iOS** In the Files app, touch and hold a file to share it, or connect the iPhone to a Mac and export the files through File Sharing in Finder.

9.4 Opening the files

The CSV files open in common analysis tools such as Excel and Python. The first line of a file is `# sample_rate=250;`; the table starts on the second line.

Python example:

```
import pandas as pd

eeg = pd.read_csv("eeg_20261023_101010_000001.csv", comment="#")
band = pd.read_csv("eeg_20261023_101010_000001_band.csv", comment="#")
print(eeg[["elapsed_ms", "ch1_uv", "ch1_clean_uv"]].head())
```

10. Settings

Change the settings on the **Settings** tab. Changes are saved immediately. **Reset to defaults** restores all settings; saved recordings are kept.

10.1 Storage

Shows the recording folder; **Open folder** opens it.

10.2 Motion detection

Setting	Description
Accel threshold	Acceleration above this value counts as movement.
Gyro threshold	Rotation faster than this value counts as movement.
Min duration	Movement lasting at least this long is marked as motion time.
Auto markers for motion	Writes markers (MOTION_START, MOTION_END) at the start and end of motion time.
Show IMU lane as x·y·z axes	Shows movement per axis.

10.3 Display

Sets the default time range and the default waveform range of the Measure screen.

10.4 Battery warnings

- **Warning**: shows a warning when the battery level is at or below this value.
- **Stop recommended**: recommends stopping the recording when the battery level is at or below this value.

10.5 Quick markers

Add, edit, and delete the quick marker names used on the Measure screen.

10.6 Language

Select the globe icon at the top right of the first screen and choose **한국어 / English / Use system language**.

11. Troubleshooting

11.1 The device is not found

- Check that the device is on and charged.
- Check that Bluetooth is on on the PC or phone.
- Check that the app has Bluetooth access (Nearby devices on Android).
- Place the device within 1 m and select **Scan again**.
- If the device is connected to another phone or PC, disconnect it there first.

11.2 The contact does not turn Good

- Check the REF and BIAS (GND) electrodes first. Poor contact of these two lowers the signal quality of every channel.
- Part the hair, add gel, and press the electrode gently in place.
- Wait 1–2 minutes for the gel to settle.
- Avoid moving, talking, and clenching the jaw during the measurement.
- Remove channels without an electrode from the layout (see 4.5).

11.3 All channels show the same noise

- Noise of the same shape on every channel usually comes from the REF or BIAS electrode or the cable. Reattach both electrodes.
- Secure the cables so that they do not swing or pull.
- Keep away from chargers, monitors, power adapters, and other electrical equipment while measuring.

11.4 ECG appears in the waveform

Sharp peaks repeating about once per second are the electrocardiogram (ECG) picked up by the electrodes.

- Make sure the REF electrode has not slipped toward the neck or body, and use the recommended position (earlobe).
- If only one channel shows it, that electrode may sit over a blood vessel. Move it slightly.

11.5 The connection drops often

- Reduce the distance between the device and the PC or phone, with no people or walls in between.
- Exclude VitalTrack from battery optimization on the phone (see 2.2).

- If many Bluetooth devices are nearby, turn off Bluetooth on those not in use.
- After reconnecting, the recording continues in the same file (see 5.3).

Appendix A. CSV columns

A.1 Measurement data (`….csv`)

Column	Unit	Content
`timestamp_utc`	UTC time	Sample time
`elapsed_ms`	ms	Time since the start of the recording (4 ms steps)
`ch1_uv` – `ch8_uv`	µV	Raw signal (as received from the device)
`ch1_clean_uv` – `ch8_clean_uv`	µV	Cleaned signal (0.5–45 Hz band-pass, mains noise removed, common reference)
`acc_x_g`, `acc_y_g`, `acc_z_g`	g	Acceleration
`gyro_x_dps`, `gyro_y_dps`, `gyro_z_dps`	°/s	Angular velocity
`acc_mag_mg`	mg	Movement magnitude from acceleration
`gyro_mag_dps`	°/s	Movement magnitude from angular velocity
`event_marker`	text	Marker names, separated by `|` when there are several

- Unused channels are stored as 0.
- Samples missing because of a dropped link are filled by linear interpolation between their neighbours; the period is marked with `SYSTEM_DATA_GAP` and `SYSTEM_DATA_RESUME`.
- Motion time is marked with `MOTION_START` and `MOTION_END`.
- Because of the hardware configuration, the raw signal is stored with inverted polarity (REF − electrode). Spectra and band power are not affected.

A.2 Band power (`…_band.csv`)

Column	Unit	Content
`timestamp_utc`	UTC time	Analysis time
`elapsed_ms`	ms	Same time axis as the measurement data; end of the 8-second analysis window

Column	Unit	Content
`ch1_delta_pct` – `ch8_gamma_pct`	%	Relative delta, theta, alpha, beta and gamma power of each channel

Channels not in the layout, channels judged Not attached, and periods before the first analysis are left empty.

Appendix B. Glossary

English	한국어	Description
Electrode	전극	Contact placed on the skin to pick up the biosignal
Channel (CH1–CH8)	채널	An input of the device; each electrode connects to one channel
Reference electrode (REF)	기준 전극	The electrode every channel is measured against
Bias electrode (BIAS)	바이어스 전극	Driven ground electrode that reduces common-mode noise; GND on the cap
Good / Weak / Not attached	정상 / 약함 / 부착 안됨	Electrode contact states
Signal preview	신호 미리보기	Viewing waveforms without recording
Marker	마커	An event name placed at a moment in a recording
Motion time	움직임 구간	Period in which the IMU detected movement
Band power	밴드 파워	Share of signal power in each frequency band
Raw signal	원본 신호	The signal as received from the device
Cleaned signal	정제 신호	The signal after filtering and common re-referencing
Sampling rate	샘플링 주파수	Measurements per second (250 Hz)

Appendix C. Specifications

Item	Specification
Channels	8 EEG channels, reference electrode (REF), bias electrode (BIAS)
Conversion	24-bit ADC (TI ADS1299), gain 24

Item	Specification
Sampling rate	250 Hz
Input range	±187.5 mV
Resolution	About 0.022 µV
Motion sensor	6-axis IMU (acceleration ±4 g, angular rate ±500 °/s, 104 Hz)
Wireless	Bluetooth Low Energy (nRF52840)
Device name	VitalTrack EEG (XXXX), XXXX is a 4-character code that differs per device
Battery	Lithium polymer 3.7 V, 750 mAh
Charging	USB-C, charging current 50 mA
Electrode connector	10 pins, 2.54 mm pitch (REF, CH1–CH8, BIAS)
Battery connector	2 pins, 1.25 mm pitch
Board size	50 × 36 mm (without the connector pins)